

Lab 3: Architectural Pattern Selection — Written Justification

Warehouse Inventory & Pallet Location Tracker

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Architecture Selection: "We chose Layered Architecture for the Warehouse Inventory & Pallet Location Tracker System."

Architectural Choice

A three-tier Layered Architecture — Presentation Layer, Business Layer, and Data Layer — was selected over Microservices and Client-Server for this system.

Reasons for Selection

- **Centralised, consistency-critical business logic:** Core requirements (FR-001 rack-capacity validation, FR-002 placement assignment, FR-004 coordinate lookup) all depend on a single, always-current view of rack occupancy and pallet locations. A layered design keeps this validation and lookup logic together in one Business Layer with one authoritative Data Layer, avoiding the data-consistency and synchronisation challenges that splitting this logic across independently deployed microservices would introduce for comparatively modest scenario complexity.
- **Right-sized for the workflow, not over-engineered:** The system has a small, well-defined set of interacting roles (Warehouse Operator, Logistics Supervisor, Scanning Device) rather than many independently-scaling client types. Layering lets the Presentation Layer (operator console, scanning interface) evolve independently of business rules, and lets the supervisor override workflow (FR-005) reuse the same Business Layer services, without the operational overhead (service discovery, inter-service network calls, distributed deployment) that Microservices would add, or the single-server bottleneck and poor fault isolation of a pure Client-Server design.

Security Advantage

Placing the Inventory & Rack Database and Audit Log Database behind the Business Layer means only authenticated Business Layer services (Movement Logging Service, Override Approval Service) are ever permitted to write to the Data Layer — no Presentation Layer component or external actor can access the database directly. This makes it straightforward to enforce a single, consistent role-based-authentication and immutable-audit-trail policy (NFR-002) for every stock-movement and placement-override record, since all writes pass through one controlled layer instead of being scattered across independently deployed services with their own access paths.

Performance Benefit

Because all pallet-location and rack-capacity data lives behind one Data Layer, the Placement Validation Service and Location Lookup Service can query a single, indexable Inventory & Rack Database directly, without the network round-trips that a Microservices split between validation, lookup, and storage would add. This keeps SKU/Pallet coordinate lookups across 100,000 bins comfortably within the sub-100ms target (NFR-001), since there is no inter-service call overhead in the lookup path.